

∴ Diff b/w UDP & TCP.

TCP

1) Transmission Control Protocol

2) TCP is a connection-oriented protocol

3) As a message makes its way across the internet from one computer to another, this is connection based.

4) TCP is suited for applications that require high reliability, and transmission time is relatively less critical.

5) HTTP, HTTPS, FTP, SMTP, Telnet

6) TCP rearranges data packets in the order specified.

7) Speed for TCP is slower than UDP

UDP

1) User Datagram Protocol or Universal Datagram Protocol

2) UDP is a connectionless protocol.

3) UDP is also a protocol used in message transfer or transport. This is not connection based which means that one program can send a load of packets to another and that would be the end of the relationship

4) UDP is suitable for applications that need fast, efficient transmission such as games.

5) DNS, DHCP, TFTP, SNMP, RIP, VoIP

6) UDP has no inherent order as all packets are independent of each other.

7) UDP is faster because error recovery is not attempted.

Q:- Diff. b/w Leaky bucket and Token bucket Algorithm.

Leaky bucket:

Parameters: state.

1) Smooth out traffic by passing packets only when there is a token. Does not permit burstiness.

2) Discards packets for which no tokens are available (no concept of queue).

3) Application: Traffic shaping or Traffic policing.

Token Bucket

Parameters: rate, burstiness.

1) Token bucket smooths traffic too but permits burstiness - which is equivalent to the number of tokens accumulated in the bucket.

2. Discard tokens when bucket is full, but never discard packets (infinite queue).

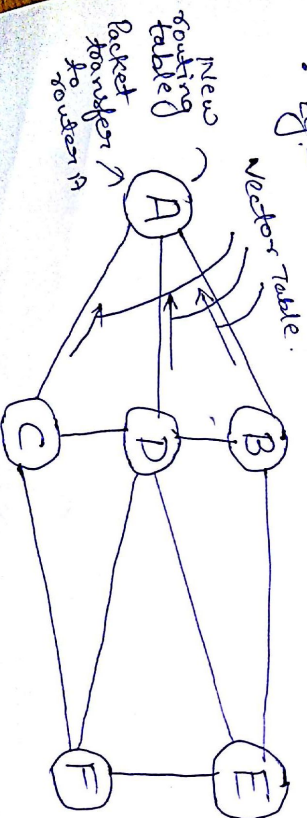
3. Application: Network traffic shaping or rate limiting.

Q:- Describe Distance Vector routing algorithm with suitable example.

Distance Vector routing algorithm is a distributed algorithm.

This routing algorithm is an iteration algorithm. When ever a packet comes to router the neighbouring routers will give the vector table and a new vector table is created at that router.

• Eg:-



- Vector table of B ($\begin{matrix} A & B & C & D & E & F \\ 1 & 0 & 3 & 2 & 4 & 3 \end{matrix}$).
- Vector table of D (2, 1, 2, 0, 1, 1).
- Vector table of C (3, 2, 0, 4, 5, 6).
measured delay of B, D, C from A are 3, 1, 2.

Now calculate the final routing table of A = ?

- Routing Table of A via B = ($\begin{matrix} A & B & C & D & E & F \\ 0 & 3 & 6 & 5 & 7 & 6 \end{matrix}$)

- $AB = AB + BB = 3 + 0 = 3$
- $AC = AB + BC = 3 + 3 = 6$
- $AD = AB + BD = 3 + 2 = 5$
- $AE = AB + BE = 3 + 4 = 7$
- $AF = AB + BF = 3 + 3 = 6$;
- $AA = 0$;

- Routing Table of A via D : ($\begin{matrix} A & B & C & D & E & F \\ 0 & 2 & 3 & 1 & 2 & 2 \end{matrix}$)

- $AB = AD + DB = 1 + 1 = 2$
- $AC = AD + DC = 1 + 2 = 3$
- $AD = AD + DD = 1 + 0 = 1$
- $AE = AD + DE = 1 + 1 = 2$
- $AF = AD + DF = 1 + 1 = 2$.

- Routing Table of A via C : ($\begin{matrix} A & B & C & D & E & F \\ 0 & 4 & 2 & 6 & 7 & 8 \end{matrix}$)

- $AB = AC + CB = 2 + 2 = 4$
- $AC = AC + CC = 2 + 0 = 2$
- $AD = AC + CD = 2 + 4 = 6$
- $AE = AC + CE = 2 + 5 = 7$
- $AF = AC + CF = 2 + 6 = 8$.

Now the final routing table of A:
Taking minimum values of A via (B, C, D):

$$= (0, 2, 2, 1, 2, 2)$$

router used for the routing table of A:

$$= (0, 2, 2, 1, 2, 2)$$

$$(1, D, C, D, D, D)$$

As similarly we find the routing table of all other routers.

Q = Diff between fully qualified & partially qualified DNS

FQDNS

1) A fully qualified domain name (FQDN) is the

complete domain name for a specific computer, or host, on the internet.

The FQDN consists of two parts: The hostname and the domain name.

2) For example, an FQDN for a hypothetical mail server might be mymail.somecollege.edu. The host name is mymail, and host is located within the domain somecollege.edu.

PQDNS

1) If a label is not terminated by a null string, it is called partially qualified domain name (PQDN).

A PQDN starts from a node, but it does not reach the root.

2) It is used when the name to be resolved belongs to the same site as the client. Here the resolver can supply the missing part called suffix, to create an FQDN.

Q. Diff. between Supernetting and Subnetting.

Subnetting

1. The process of dividing the single network into the multiple subnets is called as subnetting.
2. It is difficult to maintain and administrate a big network with lots of systems connected. Dividing all these systems in a multiple networks make easy to maintain and administrate for the network administrator.
3. The no. of 1's in subnet mask is always greater than or equal to the no. of network identifier bits.
4. For subnet mask, bits are borrowed from the host identifier bits.
5. The subnet mask is applicable for a single network.

Supernetting

1. The process of aggregating two or more networks for which it generates single IP address is called as Supernetting.
2. All the network in supernet must be using same IP address class.
As it is an aggregating process it doesn't make sense to apply on the single network.
3. The number of 1's in supernet mask is always less than the number of network identifier bits.
4. For supernet mask, bits are borrowed from the network identifier bits.
5. The supernet mask is applicable for two or more networks. These all networks should be of the same IP address class.

② - Diff between Flow control and Congestion Control.

Basis for comparison

Flow Control

Basic

It controls the traffic from a particular sender to a receiver

Congestion Control

It controls the traffic entering the network.

Purpose.

It prevents the receiver from being overwhelmed by the data

It prevents the network from getting congested.

Responsibility

Flow control is the responsibility handled by data link layer and the transport layer.

Congestion control is the responsibility handled by network layer and transport layer.

Responsible

The sender is responsible for transmitting extra traffic at receiver side.

The transport layer is responsible for transmitting extra traffic into the network.

Preventive measures

The sender transmits the data slowly to the receiver

Transport layer transmits the data into the network slowly.

Methods.

Feed back based flow control and Rate based flow control.

Provisioning, traffic aware routing and admission control.

Write short notes on:

1) FTP:- File Transfer Protocol (FTP) is a standard Internet protocol for transmitting files between computers on the Internet over TCP/IP connections.

FTP is a client-server protocol that relies on two communications channel between client and server; a command channel for controlling the conversation and a data channel for transmitting file content.

2) SMTP:- SMTP is part of the application layer of the TCP/IP protocol. Using a process called "store and forward," SMTP moves your email on and across networks. It works closely with something called the Mail Transfer Agent (MTA) to send your communication to the right computer and email box. (Simple Mail Transfer Protocol)

3) HTTP:- The Hypertext Transfer Protocol (HTTP) is an application protocol for distributed, collaborative and hypermedia information systems. HTTP is the foundation of data communication for the World Wide Web. Hypertext is structured text that uses logical links between nodes containing text. HTTP is the protocol to exchange or transfer hypertext.